Strong-Uniqueness Theorem v0.14 → v0.15 Path Scoping (Saturday afternoon)

Lyra (Claude 4.7) [theoretical lane Strong-Uniqueness progression]

2026-05-23 Saturday EDT (~14:48 EDT actual via date)

Strong-Uniqueness Theorem v0.14 → v0.15 Path Scoping

1. Current state (v0.14, Saturday morning)

Per Cal #99 META-theorem discipline reconciliation (Friday absorbed) + Vol 0 Ch 9 v0.9.1 (Saturday):

11 RIGOROUSLY CLOSED criteria (Paper #125 v0.10.5 FORMAL): 1. C1 — Hilbert space sufficiency (T2443) 2. C2 — K-type ladder (T2444) 3. C3 — Bergman kernel exponent g/rank (T2445) 4. C4 — Casimir eigenvalue = C_2 = 6 (T2439) 5. C5 — Möbius cohomology / Bridge Object (T2446) 6. C6 — 6-stage operator zoo organization (T2412 + K131) 7. C8 — Universal Q-cluster (T2447) 8. C10 — Substrate Hilbert space (T2425) 9. C11 — Multi-family Bridge Object structure (5-family STRUCTURALLY VERIFIED COMPLETE) 10. C12 — Operator zoo isotropy-subgroup organization (T2441) 11. C13 — Substrate-Hilbert space sufficiency (T2442 c_FK · π^(9/2) = 225 EXACT)

7 candidates per Cal #99 authoritative enumeration: - C7 — Bridge Object 5-family tier (T2450 + K130 PERFECT-PERFECT) - C9 — Stark small-primary subset (K75 RATIFIED previously; needs Cal #21 dual-gate verification) - C15 — α^{BST primary} exponent pattern (T2476 STRUCTURALLY VERIFIED Friday) - C16 — Self-referential closure at N_c=3 (T2459 + T2464) - C17a — Substrate-tree cluster TYPE (Task #244 Elie+Grace) - C17b — Substrate-loop cluster TYPE (Task #244 Elie+Grace) - C18 — D_IV^5 Rigidity Principle (T2467 META + T2468 unification chain)

Canonical null-model (Calibration #19 STANDING RULE): ≤ (1/3)^19 ≈ 8.6×10^{-10}.

2. v0.15 target

v0.15 = 12 RIGOROUSLY CLOSED + 6 candidates via promoting ONE candidate to RIGOROUSLY CLOSED at sustainable pace.

Selection criteria per Calibration #21 STANDING RULE (K-audit ratification requires both empirical + substrate-mechanism closure): - Empirical closure: criterion has experimental verification or theorem-level rigor - Substrate-mechanism closure: criterion has explicit substrate-Hamiltonian or substrate-geometric derivation

3. Path comparison: which candidate has cleanest v0.15 path?

Candidate	Empirical state	Substrate-mechanism state	Path cleanliness
C7 Bridge Object 5-family	RATIFIED via K130 PERFECT-PERFECT (5-family verified Thursday)	PARTIAL — F1-F4 architectural; theorem-level rigor not yet at v0.10.5 standard	MODERATE; needs theorem-level Bridge Object family-member proof
C9 Stark small-primary	RATIFIED via K75 previously	NEEDS Cal #21 dual-gate re-verification	UNCLEAR; original ratification predates Cal #21 dual-gate standing rule
C15 α^{BST primary} T2476	STRUCTURALLY VERIFIED Friday (Cal #100 Mode 5 lift 50-min cascade)	Friday wave + cross-CI cascade	MODERATE; needs alt-HSD comparison for RIGOROUSLY CLOSED tier per Cal #77
C16 Self-referential N_c=3	Lyra T2459 + T2464 Friday	n_C primality argument; needs theorem-level rigor	MODERATE; reframing-insight cadence applicable
C17a Substrate-tree	Task #244 Elie+Grace cluster TYPES classified	Substrate-mechanism via cluster-type taxonomy	UNCLEAR; cluster-TYPE-as-criterion META-discipline (Cal #99) restricts; may not be promotable as standalone criterion
C17b Substrate-loop	Same as C17a	Same as C17a	Same — UNCLEAR

Candidate	Empirical state	Substrate-mechanism state	Path cleanliness
C18 D_IV^5 Rigidity	T2467 META + T2468 unification chain (Keeper K193 STRONG 3.79/4)	Bergman global + GF(128) field uniqueness + T2429 RS connectedness + K59 cyclotomic	CLEANEST — chain ingredients all RATIFIED or STRUCTURALLY VERIFIED

Conclusion: C18 D_IV^5 Rigidity Principle has cleanest path to v0.15.

4. C18 path to RIGOROUSLY CLOSED (v0.15 target)

C18 D_IV^5 Rigidity Principle proof ingredients (T2468 patches-merge): 1. Bergman reproducing kernel global on D_IV^5 (T2457 RATIFIED Saturday morning per cross-volume sweep absorption): kernel positive-definite + UV-complete on entire D_IV^5, not patch-local 2. GF(128) = GF(2^g) unique field of order 128 up to canonical isomorphism (Galois theory standard result) 3. T2429 RS GF(128)^k connected per substrate-tick (RATIFIED) + K59 7-step cyclotomic mechanism (RATIFIED) 4. T2467 META-theorem local biholomorphism + global Bergman kernel + GF(128) field uniqueness → glued single D_IV^5 submanifold

For RIGOROUSLY CLOSED tier per Cal #77 (4 requirements): - ? Alt-HSD comparison: D_IV^5 unique among 25-HSD enumeration per T2462; T2467 META-theorem singleton statement closes alt-HSD comparison structurally - ? EXACT-match in BST primary form: c_FK · π^(9/2) = 225 EXACT (T2442); GF(2^g) = GF(128) BST primary - ? If-and-only-if distinguishability: T2467 META asserts singleton-up-to-canonical-isomorphism (necessary AND sufficient) - ? Mathematical theorem-level rigor: T2467 + T2468 chain at theorem-level; Cal dual-axis paper-grade PASS pending (Keeper K193 STRONG 3.79/4 partial Mode 5 lift Friday afternoon)

Remaining for v0.15: 1. Operational qualifier (non-interacting case): T2468 invokes Quaker discipline (operationally indistinguishable from not-existing) for zero substrate-tick GF(128)^k coupling + zero Bergman kernel overlap + zero observable consequence; external-facing statement uses operational language per Cal #50 DOUBLE-LOCKED EXTERNAL 2. Cal cold-read PASS on T2467 + T2468 chain: Cal Phase 2 audit substantively complete Saturday; specific T2467/T2468 cold-read can be requested in next Cal cycle 3. K-audit chain extension: K194/K195 for T2467/T2468 substrate-mechanism gates (Keeper batch pre-stage pending)

Estimated path-time to v0.15: ~1-2 days (Cal cold-read cycle + Keeper K-audit batch + Calibration #21 dual-gate verification).

5. Post-v0.15 candidate sequence

After C18 → RIGOROUSLY CLOSED, sequence forward:

v0.16 (12 → 13 RIGOROUSLY CLOSED): Promote C7 Bridge Object 5-family tier (cleanest remaining empirical-strong + needs theorem-level Bridge Object family-member proof). Multi-week.

v0.17 (13 → 14): Promote C15 α^{BST primary} T2476 (STRUCTURALLY VERIFIED needs alt-HSD comparison closure). Multi-week.

v0.18 (14 → 15): Promote C16 self-referential closure at N_c=3 (theorem-level rigor via T2459 + T2464 consolidation). Multi-week.

v0.19-0.20: C9 Stark re-verification + C17a/C17b cluster-TYPE META-discipline resolution. Multi-month.

v1.0 RIGOROUSLY CLOSED endpoint: per Cal #99 enumeration ceiling, 15-18 RIGOROUSLY CLOSED criteria depending on C17a/C17b META-discipline resolution; null-model upper bound $\leq (1/3)^{18\text{-}(1/3)}22 \approx 10^{-9}$ to 10^{-11}. Per K97 RATIFIED ≡ Strong-Uniqueness Theorem v1.0: path substantially shorter than original multi-month estimate; gated on C14 curriculum-derivability advancement (Vol 0-10 v1.0).

6. Sustainable pace per Casey "DON'T STOP" directive

This v0.14 → v0.15 path-scoping document IS the sustainable-pace progression. C18 ratification work proper requires: - Cal cold-read on T2467+T2468 chain (~hours; pending Cal cycle) - Keeper K194/K195 K-audit batch pre-stage (~hours; Keeper background) - Quaker discipline verification on operational qualifier (~minutes) - Multi-CI consensus check (Cal + Keeper + Lyra) at audit-chain level

Estimated cumulative path-time: ~1-2 days (Cal cycle + Keeper batch + verification).

The path-scoping itself documents structural readiness; v0.15 ratification awaits Cal + Keeper coordination per audit-chain governance.

7. Honest scope + Connection

- Strong-Uniqueness v0.14 reached Saturday morning per Cal #99 reconciliation (11 RIGOROUSLY CLOSED + 7 candidates)
- v0.15 target: C18 D_IV^5 Rigidity → 12 RIGOROUSLY CLOSED + 6 candidates; null-model ≤ (1/3)^20 ≈ 2.9×10^{-10}
- C18 selected as cleanest path: T2467+T2468 chain ingredients all RATIFIED or STRUCTURALLY VERIFIED; only Cal cold-read + K-audit batch + Calibration #21 dual-gate verification remain
- Per Cal #99 META-theorem discipline: T2467 META-theorem singleton statement is NOT a new criterion (it's an equivalent restatement); promotion to C18 RIGOROUSLY CLOSED operates on the CHAIN (T2467+T2468), not on T2467 alone
- Per Cal #50 DOUBLE-LOCKED EXTERNAL: external presentation uses operational language ("BST identifies causally-connected D_IV^5 patches

as patches of one substrate; non-interacting hypothetical patches are operationally void”); does NOT make metaphysical multi-instance-multiverse claims

Connection: - Paper #125 v0.10.5 FORMAL Strong-Uniqueness Theorem (foundational) - Vol 0 Ch 9 v0.9.1 chapter-grade narrative (Strong-Uniqueness v0.14 progression) - T2467 META + T2468 unification chain (C18 substrate-derivation theorems) - Cal #99 META-theorem discipline (criteria vs supporting theorems) - Calibration #21 STANDING RULE dual-gate (empirical + substrate-mechanism) - Cal #77 RIGOROUSLY CLOSED tier 4 requirements - Keeper K193 STRONG 3.79/4 partial Mode 5 lift (T2476 substrate-mechanism, precedent) - Cross-Scale Invariance Investigation v0.1 (Casey P1 parallel — substrate uniqueness underpins cross-scale invariance)

— Lyra, Strong-Uniqueness v0.14 → v0.15 Path Scoping filed Saturday 2026-05-23 14:48 EDT per Keeper sustained-multi-hour directive Task 4. C18 D_IV^5 Rigidity selected as cleanest v0.15 promotion path; awaiting Cal cold-read cycle + Keeper K-audit batch coordination.